MIL-H-83012C(USAF)
15 June 1987
SUPERSEDING
MIL-H-83012B(USAF)
29 August 1975

MILITARY SPECIFICATION

HAT, SERVICE, WOMAN'S (FRAME AND REMOVABLE COVER)

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the woman's service hat consisting of a frame and a detachable cover. A hat box is furnished with the hat.

1.2 Classification. The service hat shall be of the following classes, as specified (see 6.2):

Class 1 - Polyester/Wool Tropical Cloth, Blue 1608
Class 2 - Texturized Polyester Serge Cloth, Blue 1598

1.2.1 Sizes. The frame and removable cover shall be furnished in the following sizes, as specified (see 6.2):

20	21-1/2	23
20-1/2	22	23-1/2
21	22-1/2	24

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENES, Wright-Patterson AFB OH 45433-6503, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426)appearing at the end of this document or by letter.

AMSC N/A FSC 8410

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2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

L-P-504	Plastic Sheet and Film, Cellulose Acetate
V-T-276	Thread, Cotton
NN-P-71	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
DDD-L-20	Label: For Clothing, Equipage, and Tentage (General Use)
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Boxes, Shipping, Fiberboard

MILITARY

MIL-C-368	Cloth, Satin, Rayon and Cloth, Twill, Rayon
MIL-F-10884	Fasteners, Snap
MIL-P-15011	Pallets, Material Handling, Wood Post Construction, 4-Way Entry
MIL-C-21115	Cloth, Tropical: Wool, Polyester/Wool
MIL-T-43548	Thread, Polyester, Cotton-Covered and Rayon Covered
MIL-C-83448	Cloth, Serge, Polyester, Texturized

STANDARDS

FEDERAL

FED-STD-191	Textile Test Methods
FED-STD-751	Stitches, Seams and Stitchings

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Loads
MIL-STD-2073-1	DoD Materiel Procedures for Development and Application of Packaging Requirements
MS27982	Fasteners, Snap, Style 3, (Pronged Ring-Head Type)

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2.1.2 Other Government document. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

LAWS AND REGULATIONS

Rules and Regulations Under the Wool Products Labeling Act of 1939

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington DC 20402-0001.)

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia PA 19103-1187.)

THE COLOR ASSOCIATION OF THE UNITED STATES

Department of Defense (DoD) Standard Color Card of Official Standardized Shades for Sewing Threads

(Application for copies should be addressed to The Color Association of the United States, 343 Lexington Ave., New York NY 10016-0927.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Standard samples. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from the document may appear in the sample, in which case the document shall govern.

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.4, 6.2, and 6.4).

3.3 Materials (see 6.6)

3.3.1 Basic material. The basic material for the brim and the band for the class 1 hat shall conform to type III, class 4 of MIL-C-21115, USAF Color Shade No. Blue 1608. The basic material for the brim and the band for the class 2 hat shall conform to MIL-C-83448 and shall be USAF Color Shade No. Blue 1598.

3.3.2 Haircloth or cotton-nylon cloth. The material for the front stay of the brim and for the crown shall be haircloth that conforms to the requirements specified in table I or, as an alternate, cotton-nylon cloth that conforms to the requirements specified in table II.

TABLE I. Haircloth.

Characteristics	Requirements
Material identification:	
Warp	Cotton
Filling	Natural horsehair
Weight	9.7 ±0.7 ounces per square yard
Texture:	
Cotton warp	60 (two ply) per inch, minimum
Horsehair filling	90 per inch, minimum
Weave	5 harness satin, with two ends weaving as one, i.e., 2/8
Color	Natural
Stiffness:	
Warp	0.035 inch-pound
Filling	0.045 inch-pound

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broken, missing, or skipped stitches

TABLE II. Cotton-nylon cloth.

Characteristics	Requirements
Material identification:	
Warp	Cotton
Filling	Monofilament nylon
Weight	8 ±0.5 ounces per square yard
Texture:	
Warp threads	46 per inch, minimum
Filling threads	28 per inch, minimum
Weave	Plain
Color	Natural
Stiffness	0.2 load pound, minimum

3.3.3 Rayon material. The material for the top panel lining and the side panel lining of the crown shall be black and shall conform to class 1 of MIL-C-368.

3.3.4 Sweatband (ribbon). The ribbon for the sweatband shall conform to the requirements specified in table III.

3.3.5 Base fabric. The base fabric for the cover shall be a knitted cotton cloth and shall weigh 4 to 5 ounces.

3.3.5.1 Coated material. The coated material for the cover shall be a knitted cotton cloth that has been coated on the face side with a vinyl coating compound and may be flash coated on the reverse side with a vinyl coating compound. The grain shall be a simulated diamond weave embossing in accordance with the standard sample (see 6.3). The coated material for the cover shall conform to the requirements specified in table IV and the color shall conform to USAF Color Shade No. 1572 (see 6.3).

3.3.5.2 Coating compound. The coating compound for the knitted cotton cloth shall be plasticized and pigmented virgin polyvinyl chloride resin or copolymer thereof pigmented to match the color and nonluster finish of the standard sample (see 6.3). Plasticizers that are known hosts to bacteria or fungi shall not be used. The coating compound shall be nontoxic, nonirritant, and free from objectionable odor. It shall be compounded to render the finished material soft and pliable.

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TABLE III. Sweatband ribbon.

Characteristics	Requirements
Fiber content:	
Warp	Rayon
Filling	Cotton
Nonfibrous material content	No loading agent permitted
Width	7/8 ±1/16 inch
Weight	0.19 ±0.02 ounce per linear yard
Texture:	
Warp threads	122 per inch, minimum
Filling threads	30 per inch, minimum
Color	Black
Weave	Plain
pH (acidity)	5.5 to 8.5
Colorfastness:	
Crocking	Good
Wet-dry cleaning	Good
Perspiration	Good

3.3.6 Facing for cover. The webbing for facing the bottom edge of the cover shall conform to table II except that the color shall be white and the width shall be 5/8 ±1/16 inch.

3.3.7 Plastic sheeting. The plastic sheeting for the underside of the top panel lining of the crown shall be colorless, transparent, highly polished, and 0.0075 inch thick and shall conform to type II of L-P-504.

3.3.8 Felt. The felt material for reinforcing and interlining the widest portion of the circular band and for interlining the brim shall be black, shall contain 35 percent wool and 65 percent rayon, shall weigh not less than 7 ounces per square yard, and shall contain no more than 20 percent nonfibrous material. The felt material shall be treated with a durable resin so that it will retain its shape.

3.3.9 Crinoline. The crinoline for interlining the circular band, for backing the top panel lining of the crown, and for binding the felt brim shall conform to the requirements specified in table V.

3.3.10 Eyelet. The eyelet shall be brass and shall conform to the dimensions specified on figure 1. Prior to coating, the eyelet shall be

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thoroughly cleaned and degreased. The brim or flange of the eyelet shall be an enameled black or a dull black finish.

TABLE IV. Coated material.

Characteristics	Requirements
Weight	16 ±1 ounces per square inch
Bursting strength	60 pounds, minimum
Adhesion of coating	4 pounds per inch, minimum
Resistance to blocking	1/
Colorfastness:	
Crocking	Good
Light (120 standard fading hours)	Good
Water resistance	100 pounds per square inch, minimum
Flex stiffness (standard conditions):	
Wales	8.0 by 10^{-4} inch pounds, maximim
Courses	14.0 by 10^{-4} inch pounds, maximum
Flex stiffness (at -40 °C):	
Wales	25 by 14^{-4} inch pounds, maximum
Courses	51 by 10^{-4} inch pounds, maximum
Accelerated weathering	2/

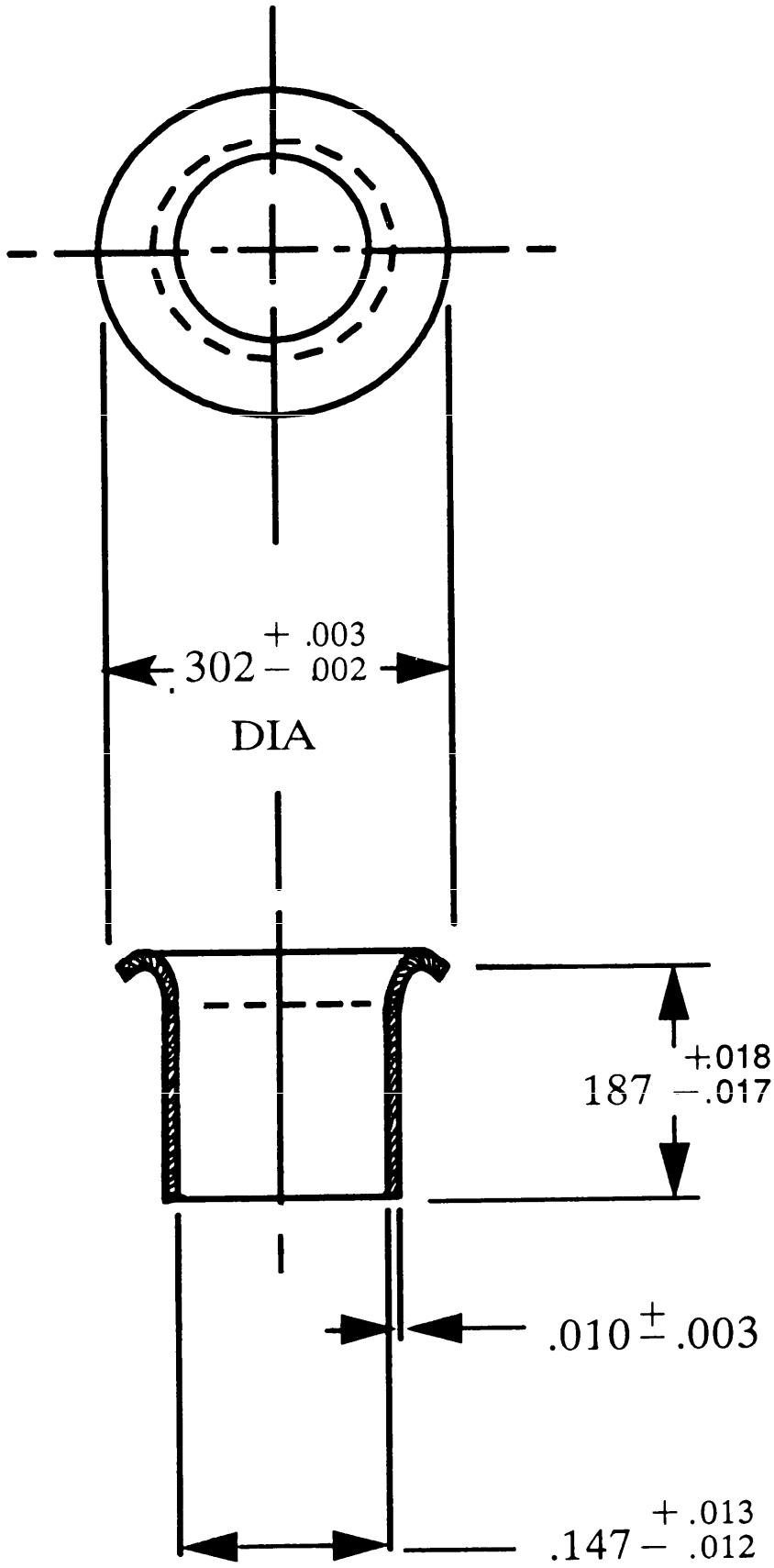
1/ A scale rating of no blocking, that is, cloth surfaces are free (see test method 5872 of FED-STD-191).

2/ There shall not be any appreciable fading of color, discoloration, exudation, or development of tackiness or stiffness.

3.3.11 Washer. The washer for backing the eyelet shall be brass and it shall fit over the barrel of the eyelet to insure a secure clinch of the eyelet.

3.3.12 Snap fasteners. The snap fasteners shall conform to style 3, finish 3 of MIL-F-10884. The components shall conform to MS27982-1N, -3N, -4N, and -5N except that the length of the head of the stud covered by MS27982-4N shall be 0.234 ±0.002 inch and the length of the eyelet covered by MS27982-5N shall be 0.335 ±0.002 inch.

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NOTE: Dimensions are in inches.

FIGURE 1. Eyelet.

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TABLE V. Crinoline.

Characteristics	Requirements
Material identification	Cotton
Color	Bleached
Weight	2.2 ±0.2 ounces per square yard
Texture:	
Warp yarns	27 per inch, minimum
Filling yarns	23 per inch, minimum
Weave	Plain
Breaking strength:	
Warp	28 pounds, minimum
Filling	18 pounds, minimum
Stiffness:	
Warp	0.016 pound
Filling	0.012 pound

3.3.13 Thread. Unless otherwise specified herein, the thread for sewing shall conform to V-T-276 as specified in table VI. As an option, thread that conforms to size 50 of MIL-T-43548 may be used. The thread for the brim shall be vat dyed and shall be Navy Blue AT, C.A. 66060 (see DoD Standard Color Card of Official Standardized Shades for Sewing Threads). The thread for the cover shall be bleached.

TABLE VI. Thread.

Color	Type	Ply	Ticket No.
Black	IA3	3	40
White	IC1	3	0
Blue	IC2	3	0

3.3.14 Labels. Each service hat shall contain a combination identification, size, personal identification and instruction label on the frame; a size label on the removable cover; and the box furnished with each hat shall contain a size label.

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3.3.14.1 Frame label. Each frame shall have a combination identification, size, personal identification and instruction label conforming to type VI, classes 4 and 10 of DDD-L-20. The label shall show fastness to dry cleaning and shall include the following information:

Name:
Service No.:
Hat, Service, Woman's
Contract No.: DLA-100-00-0-0000 (Example) 1/
Stock No.: 8410-00-000-0000 (Example) 1/
Wool Content Information 1/
Manufacturer's Name or Trademark 1/
DRY CLEAN ONLY

SIZE 1/

1/ The manufacturer shall insert the applicable information.

3.3.14.2 Removable cover label. The size label for the removable cover shall conform to type VI, class 2 of DDD-L-20 and shall show fastness to laundering.

3.3.14.3 Hat box label. The size label for the hat box shall be a removable, oval shaped, pressure sensitive adhesive label, 1-1/2 inches high by 1/2 inches wide. The label shall be white, bordered in black, and show the size of the hat contained in the hat box. The size shall be in black and shall be consistent with the height of the label.

3.4 Design. The hat shall consist of a frame and a removable white cover (see figure 2). The frame shall have a crown that conforms to the shape of the head and is made of haircloth, a blue brim that is turned down at the front and turned up at the sides and the back, a blue circular band with an integrated felt mount, and an eyelet in the center front of the circular band for attaching the insignia. An oval shaped hat box shall be furnished with each hat (see figure 3). Figure 2 is furnished solely for guidance and information. Should there be any inconsistencies between the specification and the figure, the specification shall govern.

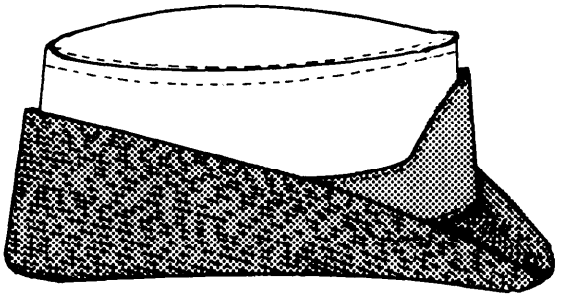
3.5 Patterns. Standard patterns will be furnished by the Government. The standard patterns shall not be altered in any way, and are to be used as a guide for cutting the contractor's working patterns. The standard patterns show size, seam allowances, directional lines for cutting, and are marked or notched for proper placement and assembly of the component parts. The working patterns shall be identical to the standard parts.

3.5.1 Pattern parts. The components of the service hat shall be cut from the materials as specified in accordance with the pattern parts indicated in table VII.

3.6 Construction

3.6.1 Hat. The construction of the service hat (frame and removable cover) shall conform in all respects to the requirements specified in table VIII and herein.

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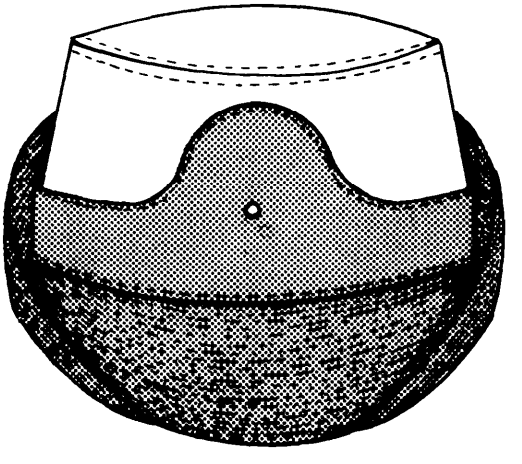


FIGURE 2. Service hat with frame and removable cover.

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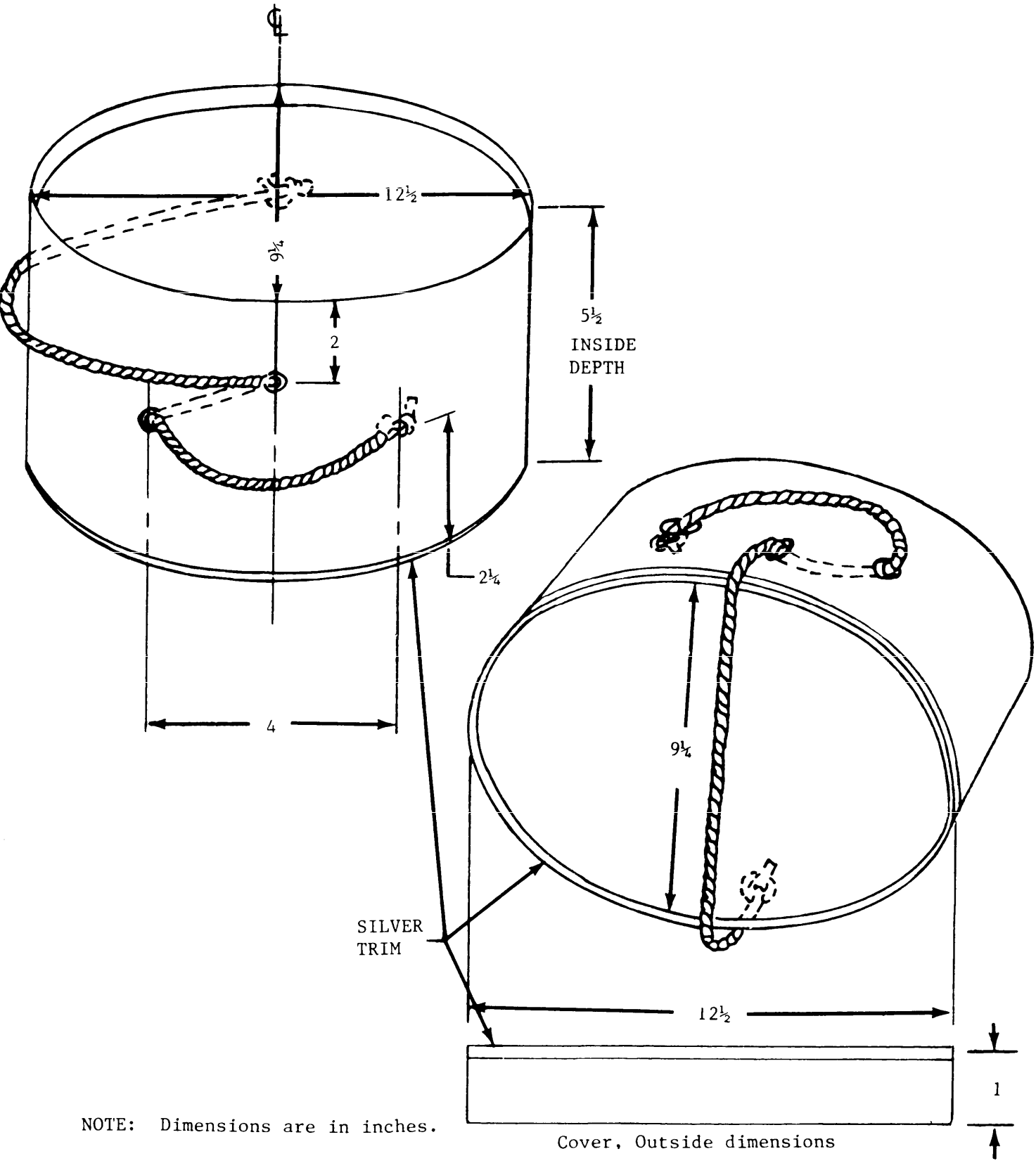


FIGURE 3. Hat box.

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3.6.1.1 Stitches, seams, and stitchings. All stitch, seam, and stitching types specified in table VIII shall conform to FED-STD-751. Whenever two or more methods of seam or stitch types are specified for the same operation, any one of them may be used. The looper (underthread) of stitch type 401 shall be on the inside of the frame.

3.6.1.2 Stitches per inch. The number of stitches per inch specified in table VIII shall be interpreted to be the minimum and the maximum number of stitches per inch permitted.

TABLE VII. List of pattern parts.

Material	Pattern nomenclature	Cut parts
Basic material (see 3.3.1)	Brim cover	2
	Hatband top piece (all sizes)	1
	Hatband under piece (all sizes)	1
Haircloth or cotton-nylon cloth (see 3.3.2)	Front brim reinforcement (all sizes)	1
	Front brim stay (all sizes)	1
	Top piece	1
	Side piece	1
Rayon material (see 3.3.3)	Side lining	1
	Top lining (all sizes)	1
Vinyl material (see 3.3.5)	Top cover	1
	Side cover	1
Plastic sheeting (see 3.3.7)	Top lining (all sizes)	1
Felt (see 3.3.8)	Brim interlining	2
	Center front piece (all sizes)	1
Crinoline (see 3.3.9)	Top interlining (all sizes)	1
	Hatband interlining (all sizes)	1
	Bias strip (foredge of felt, all sizes)	1

3.6.1.3 Thread breaks and ends of seams and stitchings. The ends of all seams and stitchings produced with stitch type 301, when not caught in other seams or stitchings, shall be backstitched not less than 1/4 inch.

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Thread breaks (all stitch types) shall be secured by stitching back of break 1/2 to 1 inch. For those seams with thread breaks, utilizing stitch type 401, a 301 stitch shall be used to secure thread breaks by stitching back of break 1/2 to 1 inch.

3.6.1.4 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks, skipped stitches, run-offs, or bobbin runouts occur during sewing, the stitching shall be repaired by superimposing and restarting the stitching a minimum of 1/4 inch back of the end of the stitching (see NOTE).

b. Except for prestitching, thread breaks, or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1/4 inch in back of the defective area, continue over the defective area, and continue a minimum of 1/4 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials, and restitching in the required manner (see NOTE).

NOTE: When making the above repairs, the ends of the repairing stitch must be backstitched and thread ends trimmed close to the hat.

3.6.2 Hat box. The hat box shall conform to the dimensions specified on figure 3. The box, including the cover, shall be made of standard commercial chipboard, completely covered on the inside with white paper and on the outside with glazed black paper. The stays around the top of the cover and the bottom of the box shall be trimmed with silver-colored paper, with 1/8 to 3/16 inch of the silver-colored paper showing on each side of the finished box. The box and cover strips shall be wire stitched. The box shall have a carrying cord made of two silver-colored strands of rayon yarn over a cotton core, twisted together to form a cord approximately 1/4 inch in diameter by 30 inches in length. The cord shall be laced through natural finish aluminum grommets on each side of the box in a manner to form a carrying handle on one side and to extend across the cover to the other side of the box in a manner that will prevent the cover from slipping off the box when it is carried. Each end of the cord shall be secured on the inside of the box by a small knot or any other means that will permit the ends of the cord to be as flat as possible against the sides of the box so that the hat will fit into the box properly. The size label of the hat box shall be located on the front of the oval, 2 inches down from the top of the cover.

3.7 Manufacturing operations requirements. The service hat shall be made in accordance with operation requirements specified in table VIII. The contractor is not required to follow the exact sequence of operations listed unless otherwise specified. Any additional basting or holding stitches used to facilitate manufacture are permissible provided the thread does not show on the finished service hat.

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TABLE VIII. Sewing operations.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
1.	Cut parts. a. Cut all parts in strict accordance with the patterns furnished, which show directional lines (warp or wale direction), size, and notches for proper assembly. b. Cut all parts made of the vinyl coated material from one piece of material. c. Cut all parts made of the basic material from one piece of material. d. Cut the haircloth for the top piece of the crown so that the straight (direction of the filling) of the haircloth will run from the center front to the center back. Cut the haircloth for the side piece of the crown so that the straight (direction of the filling) of the haircloth will run from the brim to the tip of the crown at the center front. e. If cotton-nylon cloth, in lieu of haircloth is used, cut two individual crowns and stays individually in opposite directions of the cloth. f. Cut the rayon material for lining the crown with the directional line on the pattern on the straight, in the warp direction. g. Cut the crinoline with the directional line on the pattern on the straight, in the warp direction.			

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TABLE VIII. Sewing operations. - Continued

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
2.	Replacement of damaged parts.			
	a. Replace all parts of the vinyl coated material that contain scars, scratches, spots, stains, or rough or uneven edges.			
	b. Replace any part damaged by needle chews, cuts, tears, holes, mends, or burns during manufacture.			
3.	Component marking (see 3.7.1)			
	All component parts shall be marked, ticketed, or bundled, to insure a uniform shade, size, and proper assembly throughout the service hat.			
4.	Make brim.			
	a. Stitch the ends of each ply of the felt interlining with a 3/8 inch seam allowance. Trim the seam allowance to within 1/8 inch of the seam line. Press each seam open and flat.	301 or 401	SSa-2	12-14
	b. Position the two plies of the haircloth or the two plies (opposite directions) of the cotton-nylon cloth (front stay) at the center front between the two plies of the felt interlining. Stitch lengthwise around the felt interlining, approximately 1 inch from the brim outer edge, to secure all plies.	301 or 401	LSa-1	12-14
	c. Overlap the outside edge of the felt interlining with bias-cut, 1 inch wide crinoline, and stitch so that the outer edge of the crinoline extends approximately 3/8 inch beyond the outer edge of the felt interlining.	301 or 401	LSa-1	12-14

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TABLE VIII. Sewing operations. - Continued

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
4.	Make brim. - Contd			
	d. Stitch the ends of each ply of basic material with a 3/8 inch seam allowance. Press each seam open and flat.	301	SSa-1	12-14
	e. Position a ply of basic material face up over the top surface of the felt interlining so that the outer edge of the basic material extends approximately 3/8 inch beyond the outer edge of the felt interlining and baste lengthwise 1 inch from the outer edge.			
	f. Position another ply of basic material face to face over the first ply and stitch through both plies of the basic material and the rayon lining stay piece which is around the outer edge of the felt interlining directly adjacent to the outer edge of the felt interlining with a 3/8 inch seam, without catching the felt interlining in the stitching. Trim evenly around the edge.	301	SSa-1	12-14
	g. Turn the top (second) ply of the basic material to the underside so that the felt interlining is sheathed, and baste stitch with two or three separate rows of equidistant stitching from the outer-edge to the inner edge to secure all plies of the materials properly. Trim the raw inner edges of the basic material evenly.			
	h. Stitch with continuous rows of circular stitching through all plies of the materials 1/8 inch	301	SSa-1	12-14

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TABLE VIII. Sewing operations. - Continued

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
4.	Make brim. - Contd			
	gage between stitching from the outeredge to the inner edge of the brim, removing the basting as you stitch.			
	NOTE: The completed brim shall consist of two plies of felt interlining, two plies of basic material, and two plies of haircloth or cotton-nylon cloth (front stay), in the following sequence: one ply of basic material, one ply of felt interlining, two plies of haircloth or cotton-nylon cloth (front stay), one ply of felt interlining, and one ply of basic material.			
5.	Make crown.			
	a. Overedge stitch the ends and the top of the side piece of the crown. Overedge stitch the top of the crown.	503 or 505 503 or 505	EFd-1 EFd-1	6-8 6-8
	b. Position the top piece over the side piece with the top piece overlapping the side piece 1/4 inch. Stitch 1/8 inch from the top edge. Complete the crown by overlapping one end of the side piece 1 inch over the other end, and stitch 1/2 inch from the edge. If the cotton-nylon cloth is used in lieu of the haircloth, assemble the two (opposite) crowns individually.	301 or 401	LSa-1	12-14
6.	Block crown.			
	Block and press the crown. If the cotton-nylon cloth is used in lieu of the haircloth, block the two crowns individually; then	101	EFa-1	12-14

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TABLE VIII. Sewing operations. - Continued

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
6.	Block crown. - Contd			
	block the two crowns together. Turn the edge of the crown under in accordance with the finished measurements specified herein. Stitch the folded edge 1/8 inch from the folded edge. Trim the unfinished edge of the crown close to the stitching.			
7.	Make lining.			
	a. Join the three-ply top lining (upper layer of rayon material with a backing of crinoline and an underside layer of plastic sheeting) to the one-ply side lining (rayon material), with a 1/4 inch seam, taking up the fullness of the side lining by equally distributing pleats all around. Each pleat shall be approximately 3/8 inch wide.	301 or 401	SSa-1	12-14
	b. Complete the lining by closing the back seam of the side lining.	301 or 401	SSa-1	10-12
8.	Block brim.			
	Notch the inner edge of the brim, wherever required, in accordance with trade practices. Trim evenly to obtain smooth edges for blocking. Block and press to shape; then either chalk or chalk and baste a guide line on the inner surface of the brim for use in assembling the frame.			
9.	Assemble lining, crown, and brim.			
	a. Position the lining into the crown interior, distributing the fullness around the entire circumference with the plastic top	101	LSd-1	12-14

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TABLE VIII. Sewing operations. - Continued

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
9.	Assemble lining, crown, and brim. - Contd			
	face out, and stitch the lining to the lower folded edge of the crown. Trim the excess lining.			
	b. Stitch the crown to the brim at the guide line, placing the inner edge of the brim inside the edge of the crown. The stitching shall be 1/8 to 3/16 inch from the finished folded edge of the crown. Care shall be taken to align the center front and the center back of the crown with the center front and the center back of the brim, respectively.	301 or 401	LSb-1	12-14
10.	Attach snap fastener studs.			
	Secure four studs to the face side of the crown through the brim, placing the lower edge of the stud 3/16 inch from the finished edge of the crown and as follows: one stud at the center front, one stud at each center side, and one stud 1 inch left of the center back			
11.	Attach sweatband and label.			
	Stitch the 7/8 inch wide sweatband to the inside of the brim, approximately 1/8 inch from the folded edge of the brim, catching the combination identification, size, personal identification and instruction label in the stitching at the center back (±1/2 inch) of the sweatband. Overlap the ends of the sweatband; turn the upper end back under itself. Stitch the top edge of the combination identification, size, personal identi-	301	LSa-1	12-14

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TABLE VIII. Sewing operations. - Continued

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
11.	Attach sweatband and label. - Contd			
	fication and instruction label to the ends of the sweatband 1/16 to 1/8 inch from the edge.			
12.	Make and attach band.			
	a. Position the felt center front piece on the band interlining (crinoline) at the center front and baste.			
	b. Position the interlining to the underply of the hatband basic material and stitch 1/8 inch from raw edge with the bottoms even.	301	SSe-2(a)	12-14
	c. Overedge stitch the bottom of the hatband top piece.	503 or 505	EFd-1	6-8
	d. Stitch the top ply of the basic material to the under ply of the basic material with a 1/4 inch seam.	301	SSe-2(a)	12-14
	NOTE: The circular band shall consist of the top ply of the basic material, the under ply of the basic material, the crinoline interlining, and the felt center front piece.			
	e. Turn, force out the edges, and single stitch 1/16 to 1/8 inch from the edge.	301	SSe-2(b)	12-14
	f. Fit the band to the frame, and trim the ends leaving 1/2 inch for seaming. The finished seam shall be 1-1/2 inches to the right of the center back.			
	g. Place four snap fastener sockets in the band, with the			

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TABLE VIII. Sewing operations. - Continued

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
12.	Make and attach band. - Contd			
	caps concealed inside the band, to correspond with the snap fastener studs in the frame. Exercise care in positioning the snap fastener sockets so that the lower edge of the crown will not show below the lower edge of the finished band. Insert the snap fastener sockets at the center front of the band through both the felt interlining and the under ply of the basic material. Insert the snap fastener socket at each side and at the back of the band through both the crinoline interlining and the under ply of the basic material.			
	h. Turn up the bottom of the band over the band interlining and the under ply of the basic material, and single stitch 1/16 to 1/8 inch gage from the folded edge.	301	SSbc-1	12-14
	i. Join the ends of the band at 1-1/2 inches to the right of the center back. Press the seam open and flat. The finished band shall be in accordance with the finished shaper for the band.	301	SSa-1	12-14
13.	Insert eyelet.			
	Position one eyelet, centered horizontally, in the center front of the band and 1-7/16 inches upward from the bottom edge of the finished band. Insert the eyelet through the top ply of the basic material, the felt reinforcement, the crinoline band interlining, and the under ply of the basic material; back the eyelet with a washer.			

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TABLE VIII. Sewing operations. - Continued

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
14.	Make cover.			
	a. Join the side piece of the cover at the center back with a 1/4 inch seam. Spread the seam open, and top stitch 1/16 inch gage on both sides of the seam.	301 301	SSa-1 SSz-3(b)	10-12 10-12
	b. Stitch the side piece of the cover to the top piece of the cover with a 1/4 inch seam. Spread the seam open and top stitch 1/16 inch gage on both sides of the seam. Trim 1/8 inch from the seam allowance on the top piece.	301 301	SSa-1 SSz-3(b)	10-12 10-12
	c. Block the cover and mark the finished bottom of the cover. Trim the lower edge, leaving a 3/8 inch seam allowance.			
	d. Turn in the bottom of the cover on the finished mark, and stitch the lower edge of the 5/8 inch wide webbing to the inside edge 1/16 inch from the turned-up edge. The finishing end of the webbing shall be overlapped 3/4 inch and stitched with the raw edge turned under.	301	SSb-1	10-12
	e. Stitch the upper edge of the webbing to the cover, catching the size label at the center back, ±1 inch.	301	LSa-1	12
15.	Make buttonholes .			
	Make four horizontal straight 7/8 inch long buttonholes, barred at each end, corresponding to the studs in the frame, positioned 3/8 ±1/16 inch from the bottom edge and centered at the following locations: one at the center front, one at the center	304		42 stitches per buttonhole

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TABLE VIII. Sewing operations. - Continued

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
15.	Make buttonholes. - Contd of each side, and one 1 inch left of the center back seam as worn.			
16.	Clean cover and frame. a. Trim the ends of the thread, and remove the loose ends of the thread. b. Remove all spots, stains, and shade markings without injurying the material.			
17.	Fit cover on frame. Place the cover on the crown of the frame, insuring alignment of the buttonholes with the snap fastener studs and a smooth fit. The cover shall remain on the frame.			

3.7.1 Shade and size marking. The component parts of the service hat shall be marked, ticketed, or bundled to insure a uniform shade and size throughout the service hat. Any method may be used except:

a. Corrosive metal fastening devices.

b. Sew-on shade tickets.

c. Adhesive type tickets whose adhesive causes discoloration or adheres to the material upon removal of the tickets.

NOTE: The use of an ink pad numbering machine, rubber stamp, or pencil is allowed, provided the numbering does not show on the outside of the service hat.

3.8 Finished measurements. The finished frame shall conform to the measurements shown in table IX and the finished removable cover shall conform to the measurements shown in table X.

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TABLE IX. Finished measurements (inches) of frame. 1/

Hat Size	Circumference 3/	Crown Top		Crown Sides (Heights)			Brim 1/
		Center Length (Front to Back)	Center Width (Side to Side)	Center Front	Center Sides	Center Back	Center Front Width
20	20	5-7/8	4-7/8	2-7/8	3-1/16	3-1/4	2-3/8
20-1/2	20-1/2	6	5	2-7/8	3-1/16	3-1/4	2-3/8
21	21	6-3/16	5-3/16	2-7/8	3-1/16	3-1/4	2-3/8
21-1/2	21-1/2	6-3/8	5-3/8	2-7/8	3-1/16	3-1/4	2-3/8
22	22	6-1/2	5-1/2	2-7/8	3-1/16	3-1/4	2-3/8
22-1/2	22-1/2	6-5/8	5-5/8	2-7/8	3-1/16	3-1/4	2-3/8
23	23	6-13/16	5-13/16	3	3-3/16	3-3/8	2-3/8
23-1/2	23-1/2	7	6	3	3-3/16	3-3/8	2-3/8
24	24	7-1/8	6-1/8	3	3-3/16	3-3/8	2-3/8

1/ Unless otherwise specified herein, the tolerances for the finished measurements of the frame shall be ±3/32 inch.

2/ The width of the finished brim, in its turned-up position, at the center back shall be 1/4 ±1/8 inch below the top of the crown on all sizes.

3/ The tolerance for the circumference shall be ±3/16 inch.

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TABLE X. Finished measurements (inches) of cover. 1/

Hat Size	Crown Sides		Crown Top		
	Center Front (A)	Center Sides (B)	Center Back (C)	Center Length (D)	Center Width (E)
20	3	3-3/16	3-3/8	5-7/8	4-7/8
20-1/2	3	3-3/16	3-3/8	6	5
21	3	3-3/16	3-3/8	6-3/16	5-3/16
21-1/2	3	3-3/16	3-3/8	6-3/8	5-3/8
22	3	3-3/16	3-3/8	6-1/2	5-1/2
22-1/2	3	3-3/16	3-3/8	6-5/8	5-5/8
23	3-1/8	3-5/16	3-1/2	6-13/16	5-13/16
23-1/2	3-1/8	3-5/16	3-1/2	7	6
24	3-1/8	3-5/16	3-1/2	7-1/8	6-1/8
Tolerance	±3/16	±3/16	±3/16	±3/16	±3/16

1/ The finished measurements that are designated A, B, C, D, and E shall be taken at the locations as shown on figure 4.

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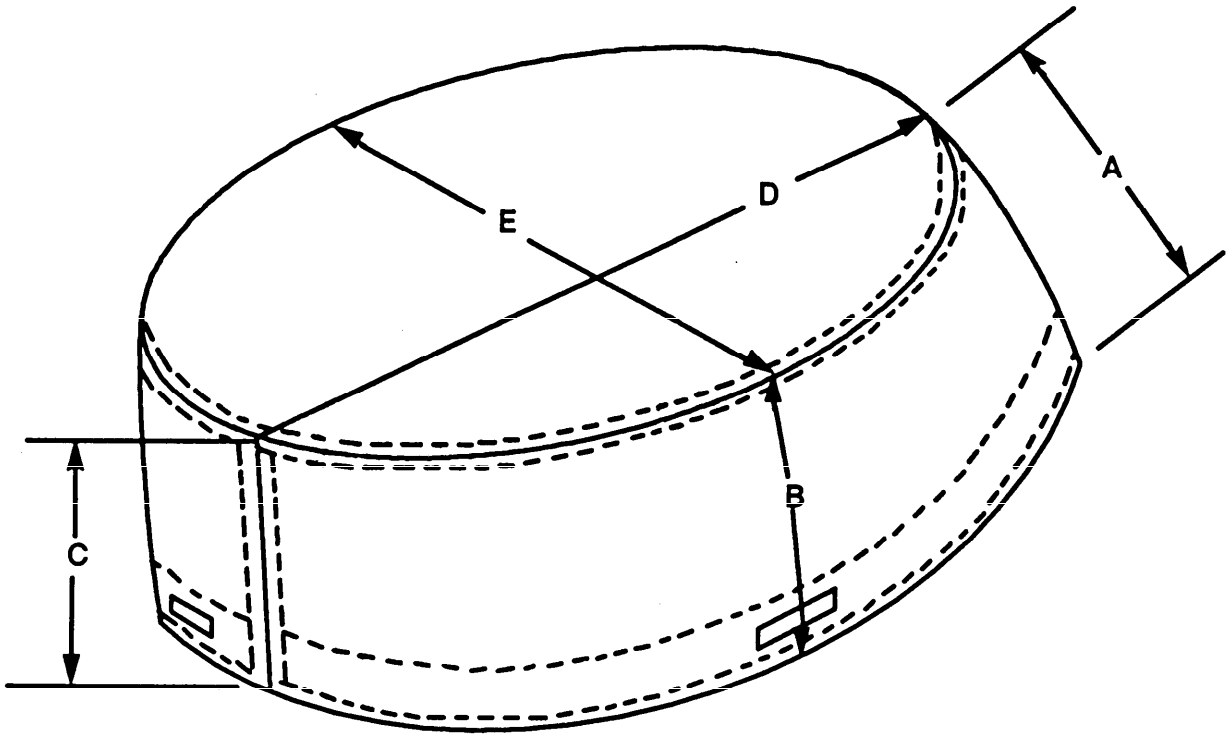


FIGURE 4. Cover (finished measurements).

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3.9 Workmanship. The finished service hat shall conform to the quality of product established by this document, and the occurrence of defects shall not exceed the acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facility suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All service hats shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with test conditions specified in FED-STD-191.

4.4 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.5.2.1 and 4.5.2.2. The presence of any defect shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be in accordance with MIL-STD-105.

4.5.1 Components and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified or qualified in this document or applicable purchase document.

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4.5.1.1 Component testing. In addition to any testing required by 4.5.1, components listed in table XI shall be tested for the for the characteristics noted. The methods of testing specified in FED-STD-191 and as listed in table XI shall be followed. A certificate of compliance shall be submitted for the requirements of the hat box (see 3.6.2). All test reports shall contain the individual values used in expressing the final results. The sample size shall be as follows:

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. The lot size units and the sample units shall be as follows:

Component	Lot size unit	Sample unit
Haircloth	Yard	1-1/4 yards full width
Sweatband	Yard	1 yard
Coated cloth	Yard	1 yard full width
Felt	Yard	1 yard full width
Crinoline	Yard	1/4 yard full width
Eyelets	Gross	15 eyelets
Washers	Gross	10 washers
Snap fasteners	Sets	20 set

4.5.2 End item examination

4.5.2.1 Visual examination. The end item shall be examined for the defects listed in tables XII and XIII. The lot size shall be expressed in units of one hat (frame and removable cover). The sample unit shall be one hat (frame and removable cover). The inspection level shall be III and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5 for major defects, 4.0 for major and minor A (combined) defects, and 10.0 for total (major and minor A and B combined) defects. Defects defined with asterisks (*) in the classification column shall be classified as major defects if the defect seriously affects appearance or serviceability. If the defect affects appearance or serviceability but not seriously it shall be classified as a minor A defect.

4.5.2.2 Dimensional examination. The end item shall be examined for conformance to the finished measurements listed in tables IX and X. Any measurement deviating from the finished dimension specified for the frame and the finished dimension specified for the removable cover shall be classified as a defect. The lot size shall be expressed in units of one hat (frame and removable cover). The sample unit shall be one hat (frame and removable cover). The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

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TABLE XI. Component tests.

Component	Characteristic	Requirement paragraph	Test method
Haircloth	Material identification:		
	Warp	3.3.2	1/
	Filling	3.3.2	1/
	Weight	3.3.2	5041
	Texture:		
	Threads per inch in warp	3.3.2	Count 1/
	Horsehairs per inch in filling	3.3.2	5202
	Weave	3.3.2	Visual 1/
	Color	3.3.2	Visual 1/
	Stiffness	3.3.2	5202
Cotton-nylon cloth	Material identification:		
	Warp	3.3.2	1/
	Filling	3.3.2	1/
	Weight	3.3.2	5041
	Texture (threads per inch)	3.3.2	5050
	Weave	3.3.2	Visual 1/
	Color	3.3.2	Visual 1/
	Stiffness	3.3.2	5202
Sweatband ribbon	Material identification:		
	Warp	3.3.4	1/
	Filling	3.3.4	1/
	Nonfibrous material content	3.3.4	1/
	Width	3.3.4	Manual 1/
	Weight	3.3.4	5041
	Texture (threads per inch)	3.3.4	1/
	Color	3.3.4	Visual 1/
	Weave	3.3.4	Visual 1/
	pH (acidity)	3.3.4	1/
	Colorfastness:		
	Crocking	3.3.4	5651
	Wet-dry cleaning	3.3.4	5622
	Perspiration	3.3.4	5680

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TABLE XI. Component tests. - Continued

Component	Characteristic	Requirement paragraph	Test method
Knitted cotton cloth	Material identification	3.3.5	1/
	Weight	3.3.5	5041
Coated cloth	Coating compound	3.3.5.1	1/
	Weight	3.3.5.1	5041
	Bursting strength	3.3.5.1	5120
	Adhesion of coating	3.3.5.1	3/
	Resistance to blocking	3.3.5.1	5872
	Colorfastness to:		
	Crocking	3.3.5.1	5651
	Light (120 standard fading hours)	3.3.5.1	5660
	Water resistance	3.3.5.1	5512 4/
	Flex stiffness at:		
	Standard conditions	3.3.5.1	5206
	-40 °C	3.3.5.1	5/
	Accelerated weathering	3.3.5.1	6/
Felt	Material identification	3.3.8	1/
	Color	3.3.8	Visual 1/
	Fiber content	3.3.8	1500 and 2101
	Nonfibrous content	3.3.8	2611
	Weight	3.3.8	5041
	Resin finish	3.3.8	1/
Crinoline	Material identification	3.3.9	1/
	Color	3.3.9	Visual 1/
	Weight	3.3.9	5041
	Texture (threads per inch)	3.3.9	1/
	Weave	3.3.9	Visual 1/
	Breaking strength	3.3.9	5100
	Stiffness	3.3.9	5202
Eyelets	Material identification	3.3.10	1/
	Finish	3.3.10	1/
	Color	3.3.10	Visual 1/
	Dimensions	3.3.10	Standard commercial 7/

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TABLE XI. Component tests. - Continued

Component	Characteristic	Requirement paragraph	Test method
Washers	Material identification	3.3.11	1/
Snap fasteners	Head length	3.3.12	Standard commercial 8/
	Post length	3.3.12	Standard commercial 8/

1/ A certificate of compliance will be acceptable for this requirement.

2/ Knitted cloth on bottom.

3/ The test method shall be 5970 of FED-STD-191, with the following exceptions: A 1-inch strip shall be tested in the wale direction only. The distance between the clamp shall be 1/2 inch at the start of the test. The period allowed for evaporation of the solvent shall be 4 hours. The testing assembly of the specimen, the plates, and the weight shall be placed in the circulating air oven for 2 hours at a temperature of 219.2 ±2 °F. The plates shall be removed, and the heating shall be continued for 4 hours at the same temperature.

4/ The water pressure shall be applied to the uncoated side of the fabric.

5/ The test method shall be 5206 of FED-STD-191, except that the specimen shall be exposed at -40 ±1 °C for a 3-hour period.

6/ The test method shall be 5804 of FED-STD-191, except that the test shall be performed for a 200-hour period on a specimen 3 inches by 6 inches (minimum) with the coated side exposed.

7/ One determination per sample unit shall be made.

8/ Five determinations per sample unit shall be made.

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TABLE XII. General defects.

Defect	Classification		
	Major	Minor A	Minor B
Material defects and workmanship damages			
Any hole or weakening defect that might develop into a hole such as a smash, multiple float, or loose slub. 1/		*	
Shade bar or unsightly slub. 1/		*	
Needle chews that might develop into a hole. 1/		*	
Scissor or knife cut, tear, or burn. 1/		*	
Any uncoated area or separation of coating. 1/	X		
Tacky, cracked, or checked coating or wrinkle or crease on coated surface. 1/	X		
Blister or lump in coated fabric. 1/	X		
Blocking and pressing			
Burned or scorched.	X		
Blocking or pressing omitted.		X	
Poorly blocked or pressed, affecting appearance.		X	
Shaded parts			
Any shaded part, clearly noticeable.		*	
Cleanness			
One or more spots or stains of permanent nature, clearly noticeable.		*	

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TABLE XII. General defects. - Continued

Defect	Classification		
	Major	Minor A	Minor B
Cleanness - Contd			
Any shade marking exposed and visible on outside of frame or cover (to be classified as defect only if clearly noticeable).	*		
One or more removable spots or stains, clearly noticeable.		X	
Thread ends not trimmed or loose thread not removed on major portion of hat.			X
One or more shade tickets not removed.			X
Component and assembly			
Any defective component (unless otherwise classified herein).	*		
Any component omitted (unless otherwise classified herein).	*		
Any required operation omitted or improperly performed (unless otherwise classified herein).	*		
Cutting			
Any part not cut in accordance with directional lines on patterns or specific requirements.	*		
Seams and stitching			
Seams twisted, puckered, or pleated, affecting appearance (unless otherwise classified herein).	*		

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TABLE XII. General defects. - Continued

Defect	Classification		
	Major	Minor A	Minor B
Seams and stitching - Contd			
Any seam open:			
- more than 1/2 inch. 2/	X		
- more than 1/4 inch but not more than 1/2 inch. 2/		X	
- more than 1/8 inch but not more than 1/4 inch. 2/			X
Any raw edge on outside:			
- more than 1/8 inch but not more than 1/4 inch. 3/			X
- more than 1/4 inch but not more than 1/2 inch. 3/		X	
- more than 1/2 inch. 3/	X		
Run-offs:			
- more than 1/4 inch but not more than 3/4 inch in edge or raised stitching but not resulting in open seam.			X
- more than 3/4 inch in edge or raised stitching but not resulting in open seam.		X	
Run-offs on joining seam. 4/			
Gage of edge, raised, or double stitching irregular, i.e., unevenly gaged or various stitching not uniform, affecting appearance.		X	
Edge or raised stitching sewn too close to edge, resulting in damage to cloth.	*		
Gage of edge, raised, or double stitching beyond range or width specified or varying more than 1/16 inch when no range is specified.			X
NOTE: The completed prim shall			

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TABLE XII. General defects. - Continued

Defect	Classification		
	Major	Minor A	Minor B
Seams and stitching - Contd			
Any part of frame or cover caught in any unrelated operation or stitching.	*		
Thread breaks or ends of all stitching (when not caught in other seams or stitching) not backstitched.		X	
Thread not specified shade or two shades of thread used on outside of frame or cover.		X	
Not specified seam or stitch type.		X	
Looper thread of stitch type 401 not on outside of finished frame.		X	
Stitch tension loose, resulting in loose seam.	X		
Stitch tension tight, i.e., stitches break when normal strain is applied to seam or stitching. 5/	X		
Stitch tension loose on edge or raised stitching resulting in loosely exposed bobbin thread or top thread.		X	
One stitch per inch less than specified on major portion of seam.			X
Two stitches per inch less than specified on major portion of seam.		X	
Three or more stitches per inch less than specified on major portion of seam.	X		
One or more stitches per inch more than specified on major portion of seam.			X

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TABLE XII. General defects. - Continued

Defect	Classification		
	Major	Minor A	Minor B
Seams and stitching - Contd			
Stitches skipped or broken on edge or raised stitching when seam is seamed, turned, and stitched:			
- more than 1/4 inch but not more than 1/2 inch.			X
- more than 1/2 inch.		X	

1/ A material defect that is weakening or nonweakening but that is inconspicuous and has a negligible effect on serviceability shall not be classified as a defect. Workmanship damage that is weakening or nonweakening but that is inconspicuous and has a negligible effect on serviceability shall be classified as a minor B defect. If the material defect or workmanship damage is definitely weakening to the material or is conspicuously located so that it is clearly noticeable, the material defect or workmanship damage shall be classified as specified herein.

2/ One or more broken or two or more continuous skipped or run-off stitches shall be classified as an open seam.

3/ A raw edge shall be classified as such only if it occurs at places where the edge is required to be turned under. A raw edge not securely caught in the stitching shall be classified as an open seam.

4/ Run-offs on a joining seam, if resulting in an open seam, shall be classified as an open seam.

5/ Puckering is evidence of tight tension. If puckering is evident, the seam shall be tested by exerting normal pull in the lengthwise direction of the seam.

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TABLE XIII. Specific defects.

Defect	Classification		
	Major	Minor A	Minor B
Eyelet			
Center of eyelet less than 1-1/4 inches or more than 1-1/2 inches above bottom edge of band.		X	
Eyelet out of alignment with center front band by more than 1/8 inch.	X		
Eyelet not inserted through all plies of band.	X		
Brass washer omitted.	X		
Eyelet and washer not securely clinched.			X
Brim			
Brim twisted, stretched, or poorly shaded, affecting appearance.	X		
Felt omitted.	X		
Haircloth or nylon-cotton cloth stay omitted	X		
Rows of quilting stitches more than 3/16 inch apart.		X	
Fullness between rows of quilting stitches.			X
Width of center front less than 2-1/4 inches or more than 2-1/2 inches.		X	
Circular band			
Crinoline interlining or felt for center front omitted.	X		
Circular band omitted, twisted, too long, too short, or poorly shaped, affecting appearance.	X		

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TABLE XIII. Specific defects. - Continued

Defect	Classification		
	Major	Minor A	Minor B
Circular band - Contd			
Circular band less than 1-1/4 inches wide on sides or back.		X	
Circular band less than 2-3/4 inches or more than 2-7/8 inches wide at center front.		X	
Sweatband			
Sweatband short, causing puckering on brim.	X		
Sweatband upper end not turned under.		X	
Excessive puckers or pleats in sweatband.		X	
Crown			
Top or side of crown twisted, poorly shaped, or crushed, affecting appearance.	X		
Puckering or pleating at joining of top piece and side piece of crown.		X	
Puckering or pleating at joining of crown side to brim.		X	
Overedge stitching (serging) of crown omitted.		X	
Crown lining			
Crown lining tight, short, or twisted, affecting appearance of outside crown.	X		
Crinoline backing on top piece lining of crown omitted.	X		
Lining exposed beyond lower edge of crown.		X	

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TABLE XIII. Specific defects. - Continued

Defect	Classification		
	Major	Minor A	Minor B
Crown lining - Contd			
Pleats at joining to top piece lining of crown not evenly distributed.		X	
Plastic piece omitted.	X		
Snap fasteners			
Any socket omitted, defective, or not securely clinched.	X		
Front socket misplaced, affecting appearance.	X		
Two or more sockets out of alignment with studs.	X		
One socket out of alignment with stud.		X	
Any stud omitted, defective, or not securely clinched.	X		
Back stud placed to right of center back.	X		
Lower edge of studs less than 1/8 inch or more than 1/4 inch from lower edge of crown.		X	
Front stud off center by 1/4 inch or more with center front.		X	
Side studs off center by 1/4 inch or more with center sides.		X	
Back stud 3/4 inch or less or 1-1/4 inches or more to left of center back.		X	
Cover			
Top piece of cover spliced.	X		

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TABLE XIII. Specific defects. - Continued

Defect	Classification		
	Major	Minor A	Minor B
Cover - Contd			
Top piece of cover not cut with straight (lengthwise) of fabric running from center front to center back.	X		
Top piece seam of cover not spread open.		X	
Top stitching omitted on one or both sides of top piece seam.		X	
Puckering of top piece at joining to side piece, affecting appearance.		*	
Top piece poorly shaped, affecting appearance.		*	
Side piece spliced.	X		
Side piece not cut with straight (lengthwise) of fabric running from lower center front to upper center front.	X		
Back seam of side pieces not spread open.		X	
Topstitching omitted on one or both sides of side piece seam.		X	
Puckered at joining of side piece to top piece or at back seam, affecting appearance.	X		
Bottom of side puckered or twisted at stitching of webbing to cover, affecting appearance.	X		
Bottom edge of side piece irregular, affecting appearance.			X
Webbing tight or twisted affecting smoothness on outside of cover.	X		

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TABLE XIII. Specific defects. - Continued

Defect	Classification		
	Major	Minor A	Minor B
Cover - Contd			
Webbing exposed beyond bottom edge of cover.			X
Buttonhole less than 5/16 inch or more than 7/16 inch from bottom edge of cover.		X	
Buttonhole not fully constructed through webbing.		X	
Center of front buttonhole off center by 1/4 inch or more with center front of cover.		X	
Center of one or both side buttonholes off center by 1/4 inch or more with center of one or both sides of cover.		X	
Center of back buttonhole 3/4 inch or less or 1-1/4 inches or more from left side of back seam of cover as worn.		X	
Center of back buttonhole positioned to right of back seam of cover as worn.	X		
Webbing not specified type, size, or color.	X		
Raw edge on end of webbing not turned under and stitched to cover.		X	
Cover placed on frame buttonholes not aligned with snap fastener studs.	X		
Crown side exposed below edge of cover, affecting appearance.	X		
Cover tight causing crown and sides of frame to wrinkle.		X	
Labels			
Label omitted or incorrect.	X		
Combination identification, size, personal identification, and instruction label not securely caught in sweatband stitching.		X	

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4.5.3 Packaging inspection. An examination shall be made to determine that preservation, packing, and marking comply with the requirements of section 5. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Examination for the closure defects shall be made on shipping containers fully packaged. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, inadequate stapling, or improper taping. Bulged or distorted container.
Content	Number of hats per container is more or less than required. Size shown on one or more hats not as specified on shipping container.

4.5.3 Palletization examination. An examination shall be made to determine that the palletization complies with the requirements of section 5. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defects
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps or film as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Commercial, as specified (see 6.2).

5.1.1 Level A. Each frame with removable cover shall be individually packaged in the oval-shaped hat box specified in 3.6.2. Neutral tissue paper shall be used to prevent shifting of the contents within the box. The box shall be further packaged individually in a standard commercial non-tear fiberboard set-up box which has dimensions not exceeding 12-3/4 inches long by 9-1/2 inches wide by 7 inches deep. The hat box shall fit snugly within the set-up box.

5.1.2 Level C. Each frame with removable cover shall be preserved in accordance with the applicable requirements of MIL-STD-2073-1.

5.1.3 Commercial. Each frame with removable cover shall be preserved in accordance with the applicable requirements of ASTM D 3951.

5.2 Packing. Packing shall be level A, B, C, or commercial, as specified (see 6.2).

5.2.1 Level A. Eight frames with removable covers, of one size and class only, preserved as specified in 5.1, shall be packed in a snug-fitting exterior-type shipping container conforming to PPP-B-601, overseas type. Closure and strapping shall be in accordance with the appendix of the container specification. Toward the end of the contract or when there are less than the quantity of the same size required per shipping container, mixed sizes (see 5.4.2) may be packed in the same container.

5.2.2 Level B. Eight frames with removable covers, of one size and class only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to type CF, class weather-resistant, variety SW, grade 200, style RSC of PPP-B-636. Closure and strapping shall be in accordance with the appendix of the container specification. Toward the end of the contract or when there are less than the quantity of the same size required per shipping container, mixed sizes (see 5.4.2) may be packed in the same container.

5.2.3 Level C. Frames with removable covers, of one size and class only, preserved as specified in 5.1 shall be packed in accordance with the applicable requirements of MIL-STD-2073-1.

5.2.4 Commercial. Frames with removable covers, of one size and class only, preserved as specified in 5.1, shall be packed in accordance with the applicable requirements of ASTM D 3951.

5.3 Palletization. When specified (see 6.2), service hats (frames with removable covers) packed as specified in 5.2, shall be palletized on a 4-way, type I, style 1, 1A, or 1B, size A, class 1 pallet fabricated in accordance with MIL-P-15011 or on a 4-way, type IV, V, or VIII, class 1, size 2, grade A pallet fabricated in accordance with NN-P-71. Each prepared load shall be in accordance with load type I or Ia and bonded with

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the applicable strap bonding or film bonding means specified in MIL-STD-147. The pallet pattern shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packages, shipping container, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Bar code markings. Bar code markings, when specified (see 6.2), shall be in accordance with MIL-STD-129.

5.4.2 Labels, mixed sizes. Each shipping container packed with mixed sizes shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label with the words "MIXED NSN'S" plainly stamped or printed thereon and under these words shall be legibly stamped or printed the correct quantity and NSN's contained therein.

6. NOTES

6.1 Intended use. The service hat covered by this specification is intended to be used as a dress hat for the women of the Department of the Air Force.

6.2 Ordering data

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class and size required (see 1.2).

c. When a first article is required (see 3.2).

d. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

e. When palletization is required (see 5.3).

f. When bar code markings are required (see 5.4).

6.3 Samples. For access to samples of the end item and shade samples, address the contracting activity issuing the invitation for bids.

6.4 First article. When a first article inspection is required, the item should be a first article sample, inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should consist of one service hat (frame with removable cover). The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and

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disposition of first article. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Subject term (key word) listing.

Hat, service
Hat, woman's

6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.3).

6.7 Changes from previous issue. Asterisks (*) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 11

Reviewer:
Air Force - 82, 99
DLA - CT

User:
Air Force - 45

Preparing activity:
Air Force - 11

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-83012C(USAF)	Hat, Service, Woman's (Frame and Removable Cover)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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